Purification of Noisy Bell States

Noise Thresholds · Physical and Virtual Protocols · Repeated Dynamics

2026.10.12 Jaehun Han, QIT Lab

Outline

1. Research motivation
2. Three protocols: 4Q, 5Q, 6Q
3. Ideal purification
4. Noisy implementation
5. Circuit decompositions
6. Equivalence of the 5Q and 6Q estimators
7. Exact Bell-diagonal maps
8. Repeated dynamics and fixed points
9. Resource comparison
10. Operational thresholds: five definitions
11. Threshold results
12. Discussion and next steps

Every number is reproduced by executed code; each claim is labelled PROVED, NUMERICAL or OPEN.

1 Research motivation

When does an *imperfect* purifier still help?

The primitive. Purification maps

$$\mathcal{P}(\rho) = \frac{\rho^2}{\mathrm{Tr}\,\rho^2},$$

concentrating weight on the dominant eigenvector and removing the leading error of a noisy Bell pair.

Two ways to use it.

- Physical — a genuine state is produced on a postselected branch. Cost: $P_{\mathrm{succ}} < 1$.
- Virtual (VD / PQEC) — only $\langle O\rangle_{\mathcal{P}} = \mathrm{Tr}(O\rho^2)/\,\mathrm{Tr}\,\rho^2$ is estimated; the purified state is *never prepared*. Cost: sampling overhead.

Where the literature stops.

Raghoonanan & Byrnes, *Quantum Error Correction by Purification*, arXiv:2603.11568 (2026): noise on the *data*, purification gadget assumed *perfect*.

This project.

Put the noise *inside the purifier* — on the Fredkin gates at the heart of the SWAP test — and ask how large the per-gate error may be before purification stops being useful.

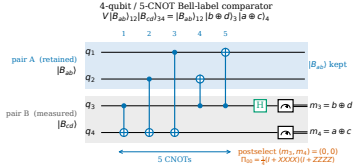
operational threshold
the largest per-CNOT noise q at which the protocol still does what it promises

Input family throughout — Bell-isotropic: $\rho(\varepsilon) = (1-\varepsilon)|\Phi^+\rangle\langle\Phi^+| + \varepsilon I/4$, $F_0 = 1 - \frac{3}{4}\varepsilon$, entangled iff $\varepsilon < 2/3$.

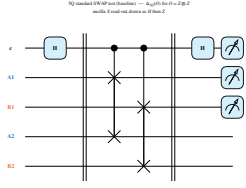
2 Three protocols

4Q global · 5Q quantum communication · 6Q LOCC

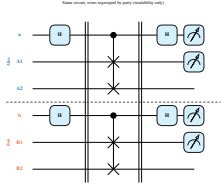
4Q postselected comparator



5Q SWAP-test PQEC



6Q distributed LOCC-VD



	qubits	ancillas	gates across the $A\|B$ cut	output	per-round cost
4Q	4	—	3 of 5 CNOTs cross	a physical state	$P_{\rm succ} = \mathrm{Tr}\,\rho^2$
5Q	5	1 shared	one ancilla must meet *both* copies	an expectation value	sampling overhead
6Q	6	1 per party	none — strictly local	an expectation value	sampling overhead

4Q needs *global* operations on one processor; 5Q needs a shared ancilla, i.e. *quantum communication*; 6Q is *LOCC* — local Fredkins plus classical post-processing.

3 Ideal purification

All three read the same object ρ^2 — but hand back different things

	what the circuit computes	what you get
4Q	$K_{00} = \sum_i \lvert B_i\rangle\langle B_i\rvert \otimes \langle B_i\rvert$: an *elementwise (Schur) square* in the Bell basis, $= \rho^2$ for Bell-diagonal ρ	a physical state $\rho_{\mathrm{out}} = \rho^2 / \operatorname{Tr}\rho^2$ on the $(0,0)$ branch, $P_{\mathrm{succ}} = \operatorname{Tr}\rho^2$
5Q	$\langle Z_a \otimes O\rangle / \langle Z_a \otimes I\rangle = \operatorname{Tr}(O\rho^2) / \operatorname{Tr}\rho^2$	a number only; the reduced state is *not* $\rho^2 / \operatorname{Tr}\rho^2$
6Q	$\Delta(O) = \langle X_a X_b \otimes O\rangle - \langle Y_a Y_b \otimes O\rangle = \operatorname{Tr}(O\rho^2)$, hence $T_{6Q} = \frac{1}{4}\sum_P \Delta(P)\, P = \rho^2$	a number only; no postselection, every shot kept

Bell-isotropic input, ideal circuit

$$\varepsilon' = \frac{\varepsilon^2}{4 - 6\varepsilon + 3\varepsilon^2}, \qquad P_{\mathrm{succ}} = \frac{4 - 6\varepsilon + 3\varepsilon^2}{4}$$

Circuit vs. independent analytics, 501 sweep points: agreement 7.8×10^{-16}. PROVED + VERIFIED

Two caveats that matter later

- the ρ^2 identity for 4Q holds *only* for Bell-diagonal ρ — otherwise a Schur square (differs by $>10^{-6}$ on 40 random states);
- 4Q is *not* LOCC, which is why it can re-entangle a separable input for $2/3 \le \varepsilon < 2 - 2\sqrt{3}/3 \approx 0.8453$.

4 Noisy implementation

Why the gate decomposition becomes part of the physics

Noise model, identical everywhere.

A two-qubit *replacement* depolarizing channel after *every* CNOT, on exactly that CNOT's wires:

$$\mathcal{D}_q^{(ij)}(\sigma) = (1-q)\,\sigma + q\left[\tfrac{I_{ij}}{4} \otimes \mathrm{Tr}_{ij}\,\sigma\right], \quad \bar{q} := 1 - q.$$

Single-qubit gates, preparation, measurement and classical post-processing stay ideal. $q = 0$ is the exact identity; $q = 1$ leaves the pair exactly $I/4$.

In the Heisenberg picture the adjoint is trivial on Pauli strings — $\mathcal{D}^\dagger(P) = P$ when $P_i = P_j = I$, else $\bar{q}P$ — which is what makes the closed forms on slide 7 possible.

A negative result that forced this model.

Depolarize at the *Fredkin* level instead (g_F on the three touched qubits):

$$F_{\mathrm{PQEC}}(p, g_F) = \frac{(1+3\alpha^2)^2}{4(1+3\alpha^4)} = F_{\mathrm{ideal}}(p), \quad \alpha^2 = 1 - \varepsilon.$$

The threshold disappears. Both parity correlators shrink by the same $(1-g_F)^2$, so the ratio is unchanged for every $g_F < 1$; the only cost is sampling, $N \sim (1-g_F)^{-4}$. VERIFIED TO 10^{-13}

Lesson. A real operational threshold needs noise that damps numerator and denominator *asymmetrically* — i.e. noise attached to the *decomposed* gates.

$\Rightarrow$ the CNOT decomposition is no longer a compilation detail. It changes the channel.

5 Circuit decompositions

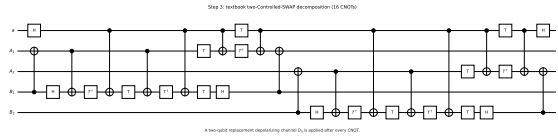
Same ideal unitary $U_{\mathrm{PQEC}} = H_a\,\mathrm{CSWAP}\,\mathrm{CSWAP}\,H_a$, three different channels

5Q	construction	CNOTs	Bell-diagonal sector	one-round loss
Type 3	textbook $\mathrm{CSWAP} = \mathrm{CNOT}\cdot\mathrm{Toffoli}\cdot\mathrm{CNOT}$ with a Clifford+T Toffoli: 8 CNOT per Fredkin	**16**	exactly invariant	$F_1 = 1 - \frac{17}{8}q + O(q^2)$
Type 4	Qiskit / pytket peephole resynthesis of the *same* unitary; $14 = 2 \times 7$, optimal per Fredkin	**14**	exactly invariant	$F_1 = 1 - \frac{7}{4}q + O(q^2)$
Type 5	gauge-matched ring ansatz learned to 3×10^{-15} at L=3 (18 CNOT), greedy-pruned to 14; **13 unreachable**	**14**	broken at $O(q)$	$F_1 = 1 - \frac{5}{4}q + O(q^2)$

The distributed 6Q versions

- **6Q-L16** from Type 3: $8 + 8$ CNOTs, 0 cross-party; each Fredkin is already local.
- **6Q-L14** from Type 4: $7 + 7$ CNOTs, 0 cross-party, but locality is *not* manifest.

Type-4 factorisation. VERIFIED 14 CNOTs split $7 + 7$ by **18 disjoint-support moves**; the blocks are gate-for-gate identical and $U_{G7} = e^{-i\pi/8}\mathrm{Fredkin}$ to 7.0×10^{-16}. Without the phase the residual is 0.39.



Type 3: the 16-CNOT round, with a depolarizing channel after every CNOT.

6 Equivalence of the 5Q and 6Q estimators

The LOCC protocol computes exactly what the shared-ancilla protocol computes

Ideal case. PROVED With $S = S_A S_B$ and R *any* four-qubit input,

$$\underbrace{C_{XX}(O) - C_{YY}(O)}_{6Q} = \tfrac{1}{2}\,\mathrm{Tr}\big[\bar{O}\,(RS + SR)\big] = \underbrace{\langle X_c \otimes \bar{O}\rangle}_{5Q}.$$

Mechanism: $X\otimes X - Y\otimes Y = 2(|00\rangle\langle 11| + |11\rangle\langle 00|)$, so the $S_A \leftrightarrow S_B$ coherences cancel and only the "neither swapped" $\leftrightarrow$ "both swapped" interference survives — precisely the single-ancilla SWAP interference.

For $R = \rho \otimes \rho$ this is $\mathrm{Tr}(O\rho^2)$; with $O = I$, the purity.

Noisy case — tested, not assumed. NUMERICAL

Raw, unnormalised Δ_5 vs. Δ_6 at identical per-CNOT noise:

	Type 3 / 6Q-L16	Type 4 / 6Q-L14
comparisons	97 440	97 440
$\max\lvert\Delta_5 - \Delta_6\rvert$	7.77×10^{-16}	5.55×10^{-16}
mean	2.1×10^{-17}	1.5×10^{-17}
$\Delta_5(I)$ vs $\Delta_6(I)$	7.8×10^{-16}	4.4×10^{-16}
$\lvert\Omega_5 - 2\Omega_6\rvert$	2.8×10^{-16}	1.7×10^{-16}
negative controls	$\geq 4.8 \times 10^{-4}$	$\geq 5.3 \times 10^{-4}$

94 two-qubit states $\times$ 56 observables $\times$ 15 values of $q \in [0, 1]$, plus 22 *arbitrary* four-qubit inputs R — the equality does not need two identical copies.

Consequence. The quantum-communication protocol (5Q) and the LOCC protocol (6Q) are the same estimator under the same noise. Going LOCC costs one extra ancilla and removes every cross-party two-qubit gate — at no statistical cost. Open: the noisy equality as a theorem.

7 Exact Bell-diagonal maps

Noisy one-round map on $\rho = \frac{1}{4}(II + xXX + yYY + zZZ)$, with $\bar{q} = 1 - q$

4Q postselected (5 CNOTs) PROVED

$$D = 1 + \bar{q}^5(x^2 + y^2) + \bar{q}^3 z^2, \qquad P_{\text{succ}} = \tfrac{D}{4}$$
$$x' = \bar{q}^3\left[(1+\bar{q})x - 2\bar{q}^2 yz\right]/D$$
$$y' = \bar{q}^4\left[(1+\bar{q})y - 2\bar{q}\, xz\right]/D$$
$$z' = \bar{q}^4\left[(1+\bar{q})z - 2\bar{q}\, xy\right]/D$$

Invariant plane $y = -z$; Bell-isotropic inputs stay on it. Matches the dense simulator to 4.4×10^{-16}.

Off-Bell sector. The exact 15×15 Jacobian at the fixed point has all twelve off-Bell eigenvalues zero — coherences enter only at second order, $\delta \to \delta^2$. A full-state attractor, not a saddle.

5Q Type 3 / **6Q-L16** (16 CNOTs) PROVED

$$D_3 = 1 + \bar{q}^4(x^2 + y^2 + z^2)$$
$$x' = \tfrac{2\bar{q}^6(x-yz)}{D_3}, \quad y' = \tfrac{2\bar{q}^6(y-xz)}{D_3}, \quad z' = \tfrac{\bar{q}^5(1+\bar{q})(z-xy)}{D_3}$$

5Q Type 4 / **6Q-L14** (14 CNOTs) — same structure, shallower exponents:

$$D_4 = 1 + \bar{q}^2(x^2 + y^2 + z^2)$$
$$x' = \tfrac{2\bar{q}^4(x-yz)}{D_4}, \quad y' = \tfrac{2\bar{q}^4(y-xz)}{D_4}, \quad z' = \tfrac{\bar{q}^3(1+\bar{q})(z-xy)}{D_4}$$

Both map $\mathcal{S}_{\text{BD}} \to \mathcal{S}_{\text{BD}}$ exactly; the plane $y = -x$ is invariant, so $u = x = -y$, $v = z$ suffice and $F = (1 + 2u + v)/4$. At $q = 0$ every map reduces to $\rho^2 / \operatorname{Tr} \rho^2$.

Type 5 does not close: one round on a Bell-diagonal input puts $O(q)$ weight on $XY - YX$, $XZ + ZX$, $YZ - ZY$ — all 15 Pauli coordinates must be tracked.

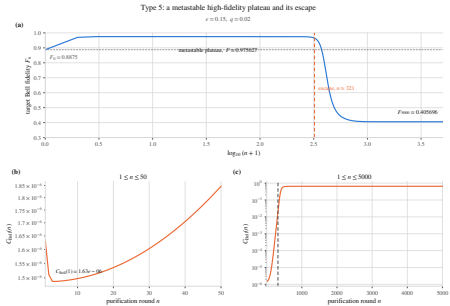
8 Repeated dynamics

$\rho_{n+1} = M_q(\rho_n)$ — bilinear in ρ, nonlinear through the normalisation

	Type 3	Type 4	4Q
CNOTs	16	14	5
$F_\star$ to $O(q)$	$1 - \frac{17}{8}q$	$1 - \frac{7}{4}q$	$1 - q$
next order	$-\frac{247}{64}q^2$	$-\frac{23}{16}q^2$	$-\frac{23}{8}q^2$
q_{SN} (fixed point exists)	0.130579	0.189417	0.180670
q_{ent} ($F_\star = 1/2$)	0.129137	0.184736	0.179815
Bell-diagonal closure	yes	yes	yes

Between q_{ent} and q_{SN} the attractor exists but is already *separable*; above q_{SN} every Bell-isotropic input decays to $I/4$.

Type 5 is qualitatively different. At $\bar{\varepsilon} = 0.9$, $q = 0.01$: $F_{10} \approx F_{500} \approx 0.987$, then $F_{1000} \approx 0.544$, $F_{5000} \approx 0.408$ — a metastable plateau, then escape ($\max|\lambda(J_{\mathrm{full}})| \approx 1.0116 > 1$) to a *product fixed circle*: purity $(1 - q/2)^2 \to 1$, but concurrence and negativity exactly zero.



Type 5: purify, plateau, escape.

Headline. The best one-round circuit is *not* the best repeated one — Type 5 has the smallest first-order loss and the worst long-run behaviour.

Scope: the Type 3/4/5 read-out is parity-*weighted*, so iterating it is an effective nonlinear map, not a shot-by-shot protocol.

9 Resource comparison

What each protocol costs, and where the CNOT savings actually come from

	qubits	CNOTs	cross-party	output	per-round cost	Bell-diag. closure
4Q comparator	4	5	3	a physical state	$P_{\mathrm{succ}} = \mathrm{Tr}\,\rho^2$	yes
5Q Type 3	5	16	—	expectation value	sampling overhead	yes
5Q Type 4	5	14	—	expectation value	sampling overhead	yes
5Q Type 5	5	14	—	expectation value	sampling overhead	no
6Q-L16	6	16	0	expectation value	sampling overhead	yes
6Q-L14	6	14	0	expectation value	sampling overhead	yes
destructive VD (reference)	4	2	—	$\langle O \rangle$ per measurement setting	many settings	n/a

Where the savings come from

circuit must reproduce	min. CNOTs
the full unitary U	**14** (13 unreachable)
the ancilla-$\|0\rangle$ isometry	**14** (no gain)
the observable, parity read-out	**5**
the observable, destructive read-out	**2**

Every saving comes from relaxing the observable; relaxing unitary to isometry buys nothing.

One-round threshold ε_2^* — placement, not count

input ε	Type 3 (16)	Type 4 (14)	Type 5 (14)	destructive (2)
0.10	0.033	0.041	0.060	0.146
0.40	0.103	0.140	0.237	0.313
0.60	0.126	0.178	0.338	0.343

16 $\rightarrow$ 14 buys 1.25–1.42$\times$, more than the ratio 1.14; two *exact* 14-CNOT implementations differ by 1.5–1.9$\times$. $\Rightarrow$ **count, depth and noiseless fidelity do not characterise a noisy PQEC circuit.**

10 Operational thresholds

Five inequivalent questions hide behind “the threshold”

Target Φ^+; $p_i^{(n)} = \langle B_i|\rho_n|B_i\rangle$, $R_i^{(n)} = p_i^{(n)}/p_{\Phi^+}^{(n)}$, $F_n = \langle\Phi^+|\rho_n|\Phi^+\rangle$.

		predicate	what it certifies
1	q_{abs}	$p_i^{(n+1)} < p_i^{(n)}$ for every non-target i and every $n \geq 0$	every Bell error population falls
2	q_{rel}	$R_i^{(n+1)} < R_i^{(n)}$ for every non-target i and every n	errors fall *relative* to the target
3	q_{th}	$F_{n+1} > F_n$ for every $n \geq 0$	the fidelity really improves
4	q_{fixed}	the target fixed point exists (saddle node, discriminant $= 0$)	repetition converges somewhere useful
5	q_{target}	the target fixed point still has $F_\star > 1/2$	the limit is still entangled

Prior work’s criterion

$F_{\mathrm{out}} > F_{\mathrm{bare}}$, *one* round, noise on the data only — the $n = 0$ case of (3), kept separate as a diagnostic:

$$q_{\mathrm{th}}^{\mathrm{first}} = \sup\{q : F_1 > F_0\}$$

$$\text{vs.} \quad q_{\mathrm{th}}^{\mathrm{full}} = \sup\{q : F_{n+1} > F_n \ \forall n\}.$$

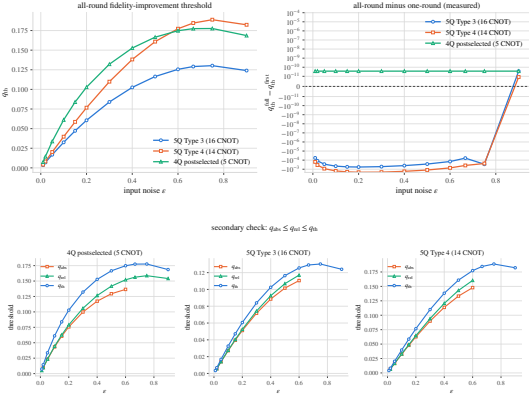
Whether the two coincide is an *output*, not an assumption.

Rules we imposed on ourselves

- measured from the *full* noisy circuit: no analytic recurrence, no closed-form map, no invariant-plane projection, no twirling;
- no previously measured threshold used as a bracket or hint;
- two independent implementation routes (agreeing to 1.7×10^{-15}) plus a 400-bit ball-arithmetic route;
- orderings and rankings compared only *after* the primary tables were frozen.

11 Threshold results

Circuit-level numerics: 39 (protocol, ε) points, 37 bisected to 7.45×10^{-11}



(a) The hierarchy holds at all 39 points NUMERICAL

$\varepsilon = 0.40$	q_{abs}	q_{rel}	q_{th}
Type 3	0.08857	0.09253	0.10255
Type 4	0.11364	0.12058	0.13815
4Q	0.11740	0.12651	0.15269

(b) All-round vs. one-round — protocol dependent

- **4Q**: equal at every ε to 7.45×10^{-11}; binding round 0.
- **Type 3** / **4**: strictly smaller, by 6×10^{-5} . . . 2.2×10^{-3}; the binding round grows with ε to 28 and 36.
- **107** trajectories rise, then turn over; **0** do the opposite.

(c) The ranking crosses between $\varepsilon = 0.50$ and 0.60

4Q > T4 > T3 $\longrightarrow$ T4 > 4Q > T3. Maxima near $\varepsilon = 0.75$: 0.1888 / 0.17759 / 0.13024.

Caveat. The Type 3/4 all-round threshold is set by a *late-round* dip, so it carries a tolerance dependence of 10^{-7}–10^{-6} — digits past the sixth are not physical. The gap to $q_{\mathrm{th}}^{\mathrm{first}}$ is 10^2–10^4 larger, so the inequality stands.

12 Discussion and next steps

What is settled, what is open, what the paper needs

Settled

- exact noisy Bell-diagonal maps and fixed points for 4Q, Type 3 and Type 4, with q_{SN}, q_{ent} in closed form;
- Fredkin-level depolarizing is self-mitigating — the threshold lives in the *decomposition*;
- the 14-CNOT Type-4 circuit factorises into two $e^{-i\pi/8}\times$Fredkin blocks, making 6Q-L14 well defined;
- 5Q $\equiv$ 6Q as estimators: proved ideally, 10^5 comparisons at 10^{-16} under noise;
- $q_{\mathrm{th}}^{\mathrm{abs}} \le q_{\mathrm{th}}^{\mathrm{rel}} \le q_{\mathrm{th}}$ at every tested point, and $q_{\mathrm{th}}^{\mathrm{full}} < q_{\mathrm{th}}^{\mathrm{first}}$ for Type 3/4.

Open — analytic

- a closed form for $q_{\mathrm{th}}(\varepsilon)$, for any protocol;
- why the all-round and one-round thresholds coincide for 4Q and strictly do not for Type 3/4; where the binding round returns to 0 (the $\varepsilon = 0.75$ rows are unresolved);
- the noisy 5Q $\equiv$ 6Q equality as a theorem;
- a symbolic proof of the Type 3/4 two-sided Pauli covariance for all q (checked at four values so far).

Open — Type 5 and the full state

- a fully symbolic derivation of the product fixed *circle* (now semi-analytic, $\|M(\rho_\star) - \rho_\star\|_F \sim 10^{-10}$);
- a complete classification of fixed points and basins — the circle is only the observed asymptotic family;
- is it the R_X, R_Y, R_Z layers themselves that break the symmetry, or only this learned placement?

Next circuits and noise models

- Cruz–Murta 7-CNOT Fredkin (arXiv:2305.18128) and Smolin–DiVincenzo five two-qubit gates (PRA **53**, 2855) in the same CNOT-only model;
- coherent and biased CNOT errors;
- entanglement-based thresholds (concurrency, negativity) and their multi-round accumulation;
- 6Q-L14 / 6Q-L16 under *asymmetric* party noise.

For the paper freeze the protocol table and the three threshold tables; promote the two 10^{-16} equivalences to lemmas; one figure set, one reproducibility appendix.